A Study of Weight-Space Programming for Robotic Manipulation Policies

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Abstract

Weight-space programming (WSP) uses a language model to inspect and modify another model's parameters. We review early evidence and test repair, iteration, and preservation in one $\pi_{0.5}$ stacking task. Across 1,056 simulated episodes, five searches produced four nonzero edits and eleven identity proposals; none of the edits passed the frozen gate. A separate fixed-batch16 follow-up validated all 80 requested activation items but yielded no accepted repair. Held-out unchanged-policy comparisons disagreed on 37/64 and 29/64 outcomes. All four edits changed fixed-input outputs without establishing closed-loop improvement. Replay failures, incorrect diagnostic annotations, a restricted interface, and a preservation gate below observed repeat variability limit the negative conclusion. We argue for evaluating policy family and failure mechanism separately: localized control defects may be more amenable to repair than perceptual ambiguity or missing competence, but this remains an untested comparative hypothesis here. Targeted demonstrations establish promise, not routine, reliable policy improvement. Physical AI adds action-dependent observations, contact dynamics, and expensive behavioral validation. Our results motivate calibrated controls and matched reproductions, rather than a general verdict on WSP.

1 Introduction

Weight-space programming (WSP) asks a general-purpose language model to reason about a network and construct parameter changes directly [1]. This offers a model-maintenance interface, while still requiring diagnostic evidence, evaluation, and previously learned competence.

For Physical AI, the target policy and failure mechanism matter. State-input controllers, vision-language-action (VLA) policies, and world-action models (WAMs) expose different representations and diagnostic interfaces. A mistimed motor command, ambiguous visual input, and an absent skill need not admit the same repair. Edits also change contacts, future observations, and subsequent actions. Small parameter support therefore guarantees neither small behavioral impact nor inexpensive validation. These distinctions motivate hypotheses, not an established ranking of policy families.

Our 1,056-episode study tests *bounded, feedback-assisted repair* of one VLA, separating development selection from held-out controls. Together with the review, it examines the conditions under which an enhancement loop could support credible improvement.

2 Early Literature Review

WSP and model editing. Takehi et al. [1] demonstrate weight interpretation, policy construction, and localized repair, including a small simulated car-control policy and a looping language model. Their non-looping-model follow-up and independent baseline audit [2] motivate testing repair separately from new competence and comparing against well-configured unedited systems. Parameter editing itself has substantial precedent: ROME targets factual associations [3], while task arithmetic combines differences obtained through fine-tuning [4]. WSP's distinctive element is a general-purpose reasoning model choosing and constructing interventions, rather than parameter modification alone.

Scope and evidence. Table 1 covers the experimental reports, replications, proposals, infrastructure, and critiques identified in our November 2–4 review, with a November 5 addendum for Guo and Xie and our own results. This is a rapid, source-based review, not an exhaustive systematic search or an independent reproduction. "Artifact-backed" means code, logs, or detailed records were inspected; it does not certify the result. The table separates intervention, reported benefit, and limitations. Missing controls are unverified evidence, not proof that a method fails; headline gains are not pooled across protocols.

Table 1: Critical evidence map: November 2–4 review, November 5 addendum. Benefits are source-reported except our own results; protocols and denominators differ. A: artifacts inspected; R: first-person report; P: proposal; C: critique; S: this study. These labels are not quality scores or independent replication. Missing reporting is not proof of a missing control.

Work / evidence	Intervention / scope	Reported benefit / finding	Limitations / interpretation
Takehi et al. [A] [1]	Read, construct, and repair networks.	Car 0/256 to 1,024/1,024; Qwen loops repaired.	One car level; non-looping Gemma gain unestablished.
Khaliq et al. [A] [11]	Rebuild and hash-check Qwen patch.	MATH-500 51.0%, four samples/problem.	Only 25 baseline problems; artifact reproduction, not discovery.
Toledo [A] [2]	Patch, grading, and baseline audit.	Unedited Qwen reaches 44.0% with changed settings.	Not a matched comparison to the original 48.1%; does not negate fixed-setting gains.
leofmarciano [A] [12]	Opus scales four Qwen neurons.	Independent thinking-mode improvement.	No consistent non-thinking gain; fresh-data benefit weakens.
Willison [A] [13]	Other models read small-policy prompts.	Qwen identifies three of four games.	Misses Freeway; reading is not repair or large-model understanding.
Krentsel [R] [14]	Single-response Breakout construction.	Reports construction across model generations.	One game; plotted scores not verified in this review.
Galatolo [A] [15]	Classify anonymized control networks.	Three of four correctly identified.	One confident confusion; no policy repair.
Singh [A] [16]	959 programs/13 tasks; 420 edit predictions.	Ten algorithmic tasks solved.	Architecture changes; language-data leakage; modest weight-access benefit.
Dolci [A] [17]	Fitted activation direction; tense steering.	Nonpreferred tense rates decrease.	Some controls break; ridge fitting; competitive LoRA baseline.
nutanc [R] [18]	Translation ablations; story activation shift.	Repetition 32% to 2% on 100 prompts.	Translation repair fails broadly; markup worsens; runtime hook, not saved weights.
Nusrat & Nusrat [A] [19]	Audit small synthetic decision networks.	16/18 planted defects detected and repaired.	Only 2/5 healthy networks cleared; simple baselines compete; no clinical inference.
Gauthier [A] [20]	Quantum-model transfer; code/weight edits.	Strong results after further training.	Editing alone trails references; fine-tuning and Lanczos contribute.
nitayrabi [R] [21]	Coding-model and harness changes.	Reported successes 16/23 to 25/27.	Different denominators; concurrent template fixes defeat weight-only attribution.
Zheng & Doubao [P] [22]	Restoration diffusion edit loop.	Conceptual iterative-repair proposal.	No measured restoration gain; not an empirical success.
WSSE / wpgit [A] [23]	Versioned patches and regression gates.	Composition/rebasing infrastructure.	Edits can conflict; some repairs train; no robot efficacy result.
Look [R] [8]	Edit UniAD driving checkpoint.	L2 0.942 to 0.845m; collision 0.29% to 0.17%.	Open-loop, mostly startup; closed-loop validation pending.
Gupta et al. [R] [5]	Ten-task neuron edits; folding task vector.	Direct gains on 5/10 tasks; folding 51% to 75%.	Folding uses trained differences; its final denominator unspecified; task-specific gains.
Guo & Xie [R] [6, 7]	Feature/attention probes; VLA, WAM, G1 edits.	Five-task VLA 27% to 65%; one task 28/92 to 82/92.	Task-fine-tuned models; cross-task retention untested; patch/logs not located.
Honnibal; Goedecke [C] [24, 25]	Protocol and weight-statistic critiques.	Distinguish reading, construction, and repair.	Methodological critique, not a universal failure result.
OpenRUA [A, adjacent] [9]	Robot-use harness; weights unchanged.	Performance varies with tools/observations.	Harness evidence, not direct weight editing.
ZAP [A, adjacent] [10]	Simulation-derived supervision and RL.	Reports real-robot transfer.	Training is central; predates WSP, not its replication.
This study [S]	One task; five searches; four actual edits.	Fixed-input effects; no accepted repair.	Noisy repeats, flawed annotations, narrow interface; not a general WSP refutation.

Implications for manipulation. Gupta et al. report direct-edit gains on five of ten tasks, separately from a training-derived folding direction [5]. Guo and Xie describe feature/attention probes followed by sparse edits of task-fine-tuned policies [6, 7]; their claimed within-task scene transfer is not cross-task preservation, and no matching artifacts were located. Look's driving metric is open-loop [8]; OpenRUA and ZAP concern harnesses and training [9, 10]. These reports motivate separating policy family, failure mechanism, and editor access in controlled comparisons.

3 Methods and Evaluation Protocol

3.1 Policy and editing interface

Let $\pi_\theta(a_{t:t+H-1} \mid o_t, \ell, \xi_t)$ produce an action chunk from observation o_t, instruction ℓ, and sampling seed ξ_t. Our target is the DROID joint-position checkpoint of $\pi_{0.5}$ [26]: two 224×224 RGB views, seven joint positions, and a gripper state yield $H = 15$ actions with eight used channels. RoboLab executes the chunks at 15 Hz with a 900-step horizon. We evaluate only StackYellowOnRedTask, with the fixed instruction "Stack the yellow block on the red block". Success uses the installed contact/geometry predicate; it does not additionally certify gripper release or sustained settling.

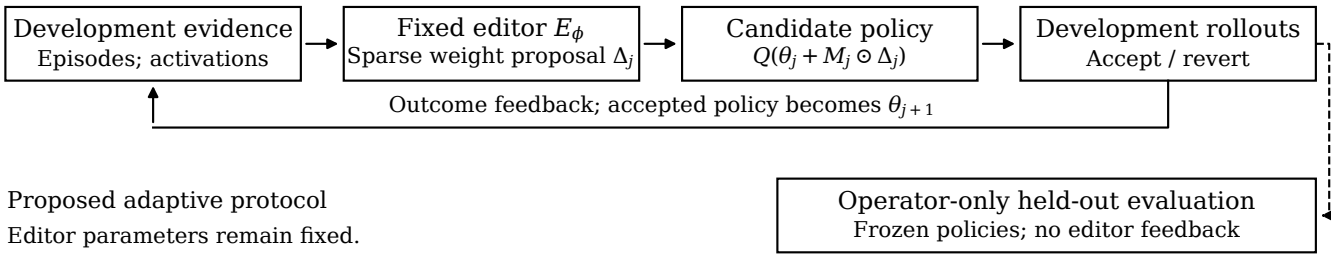


Figure 1: Proposed enhancement loop. Development feedback can change the next proposal; held-out outcomes cannot. The large editor remains fixed while the smaller policy changes.

An editor E_ϕ proposes Δ_j from development evidence and prior feedback:

$$\tilde{\theta}_{j+1} = Q_{\text{bf16}}\left[\theta_j + M_j \odot E_\phi(\theta_j, \mathcal{D}_{\text{dev}}, \mathcal{H}_j)\right]. \tag{1}$$

Here M_j restricts entries, Q applies deployment precision, and $\mathcal{H}_j$ records feedback. The gate accepts or reverts; editor weights ϕ remain fixed. Statistics, contrasts, projections, and tensor operations are permitted; optimization, regression fitting, policy-code/prompt changes, and training-derived task vectors are excluded. Gripper calibration is a separate runtime control.

We expose 18 action-expert down projections of shape 4096×1024. Each row maps one hidden-unit activation into the residual stream. A unit direction v from a success/failure residual contrast can define $w_i' = w_i - \alpha\langle w_i, v\rangle v$, a direct operation rather than a fitted update. Interpretation depends on activation timing and later nonlinear computation; contrasts may reflect consequences rather than causes of failure.

3.2 Three research questions

RQ1: Can sparse repair improve outcomes? Compare stacking edits with the base and gripper calibration; report success, nominal fixes/breakages, cost, and no-edit abstentions.

RQ2: Does adaptive iteration help? Compare three feedback-guided proposals with three base-anchored proposals committed before feedback. Evidence starts identically and budget caps match; feedback timing and usability differ. Retain all proposals and incumbents.

RQ3: What behavior is preserved? Count nominal fixes/breakages and repeat the base. This one-task study varies policy-sampling streams with environment seed zero; it tests neither new layouts nor cross-task preservation.

3.3 Budget, controls, and analysis

Four Astra (gpt-6-astra, maximum effort) runs cross two restarts with two proposal modes. Each receives 32 common base episodes, 96 candidate episodes, 96 logical activation items, and two hours. Cumulative support is limited to 128 rows in two layers (131,072 entries); masks, deployed bfloat16 changes, norms, and hypotheses are recorded.

Acceptance and attribution. For paired development outcomes $b_s, c_s \in \{0, 1\}$, define nominal fixes $F = \sum_s (1 - b_s)c_s$ and breakages $B = \sum_s b_s(1 - c_s)$. Eligibility requires a deployed-value change, more successes than the base, and $B \leq 2$. Rank eligible proposals by successes, smaller cumulative delta norm, then earlier order; otherwise retain the base. This engineering gate is not a non-inferiority test. Identity proposals consume evaluations but are ineligible. Accepted scores are nondecreasing by construction; retained points reuse observations.

Development and operator-only test slots are disjoint. A prospective v1.1 amendment precedes held-out outcomes and preserves discovery. Policies are grouped by actual bfloat16 values, checkpoint, runtime, task, prompt, environment settings, and threshold. Each unique edit, identity, and runtime control receives 64-base/64-candidate episodes; an extra 64-base repeat reuses the identity reference. Two-arm comparisons use ABBA32 order; global order is also fixed, not randomized. Round labels share estimates. Nominal pairing ensures neither equal observations nor variance reduction.

Report $\widehat{\delta} = (F - B)/n$ with nominal 95% percentile intervals from 10,000 episode-slot bootstrap resamples, separately by restart. Wilson intervals are descriptive; uncorrected McNemar tests are exploratory. Shared 16-environment execution, reused streams, and selection limit independence. Two development blocks and two restarts cannot establish reliable cluster-adjusted inference or editor variability; no confirmatory significance, generalization, or non-inferiority is claimed.

Table 2: Measured unedited-policy evidence. Development calibration is a runtime control, not WSP.

Quantity	Observation
Historical smoke: successes / episodes	9 / 16
Fresh development base ($\tau = 0.5$)	11 / 32
Runtime control ($\tau = 0.4$)	18 / 32
Runtime control ($\tau = 0.6$)	15 / 32
Identity replay: exact / checked	3 / 3
Historical-action replay: max. error	0.00683

Table 3: Final controls: 64 paired slots each. F/B: nominal fixes/breakages. *Reuses the original identity reference.

Control	Base / candidate	F / B
Fixed16 identity	24 / 27	16 / 13
Original identity	26 / 23	17 / 20
Original gripper 0.4	23 / 36	21 / 8
Original repeat*	26 / 35	21 / 12

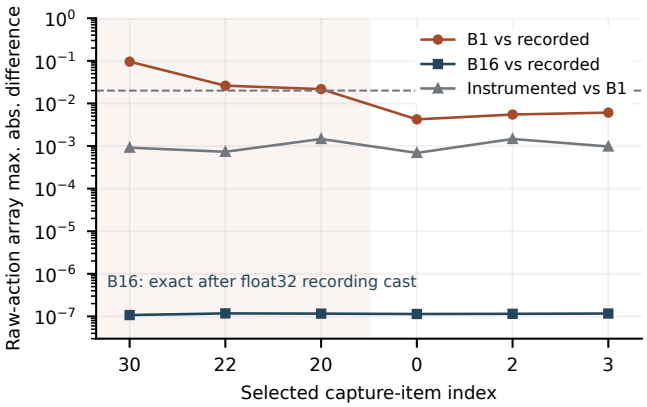


Figure 2: Six requests from five episodes: three largest failed-capture errors (shaded) and first three passing items. Raw-array errors precede the recording cast; B16 is exact at float32 precision. Dashed line: fixed 0.02 gate. This selected sample does not estimate error frequency.

Original budgets are 96 base/calibration, 384 candidate, and $320 + 128U$ final episodes for U edited incumbents. Reserved blocks and failures remain charged; no silent retries or imputation.

Separate exploratory v2 follow-up. A separate pre-held-out protocol specifies three adaptive 32-episode proposals, 96 activation items, unchanged support/gate, a fresh 32-episode baseline, and disjoint test slots. Inference pads packets to 16; complete membership, positions, seeds, and outputs support replay without skips. Features widen bfloat16 to float32 (v1: float16); raw bits and padded costs remain recorded. Final scoring uses 128 episodes per unique incumbent or identity, with shared round labels. Searches freeze before held-out feedback; declared execution is v2 then v1.1, with ABBA32 comparisons. Changed execution, precision, and baseline preclude pooling or causal version rankings. Full protocols and hashes accompany the blog.

A model-neutral handoff supports optional matched Opus trials; none is reported. Policy-family and motor/perception comparisons are future extensions, outside these frozen RQs.

4 Empirical Results and Engineering

4.1 Development controls and numerical replay

The historical smoke scored 9/16; the separate original development baseline scored 11/32. Runtime gripper thresholds 0.4/0.6 scored 18/32 and 15/32, with nominal fix/break counts 14/7 and 9/5; selection chose 0.4. These controls establish neither WSP gains nor a motor-versus-perception advantage.

Identity loading verified all 18 slices and reproduced three actions exactly; historical error was at most 0.006835. A later 32-item single-request capture failed parity (0.09474); four- and 16-item captures passed. Failed arrays remained archived and accessible, but an external audit excluded their use in the reported edit.

On six selected requests (Figure 2), native B16 reproduced historical chunks after the recording cast; B1 differed by up to 0.09566, with instrumentation adding at most 0.001465. Repeat/identity controls were exact. B16 duplicated one input, not the original mixed packet. No threshold-0.5 gripper flips occurred in 90 action rows. Batch sensitivity here establishes neither its frequency nor closed-loop effects; mixed joint/gripper errors have no single physical unit.

4.2 Four original discovery runs

All four runs retained the base policy (Table 4; Figure 3). Each run evaluated three proposals; only three of these twelve proposals changed weights. Three nonzero proposals changed 7/2/1 layer-17 rows and 4,412/1,543/791 deployed values, with ℓ_2 norms 0.02229/0.007959/0.006117. They scored 12/32, 13/32, 13/32, with nominal fixes/breakages 7/6, 9/7, 7/5; each failed preservation. The first projected selected rows away from a passing success-minus-failure activation contrast. A plausible contrast did not establish selective repair.

Nine identity proposals scored 8–20/32. The 20/32 identity cohort had a nominal paired difference 0.2813, with

Table 4: Original discovery, each against the 11/32 baseline. A/N: adaptive/non-adaptive; 32 episodes per proposal. Nonzero counts proposals that changed deployed weights, not iterations or individual parameters. All runs have three evaluations; other proposals are identities. Capture items belong to validated (V), failed (F), or unchecked (U) whole calls. Min.: driver wall minutes, including reasoning, tools, simulation and waiting.

Run	Successes: p1 / p2 / p3	Nonzero	Capture items: V / F / U	Min.	Finalist
A1	12 / 14 / 20	1	20 / 32 / 28	61.6	base
N1	13 / 17 / 10	0	0 / 96 / 0	50.9	base
A2	13 / 14 / 13	2	4 / 92 / 0	67.3	base
N2	12 / 8 / 12	0	0 / 96 / 0	47.6	base

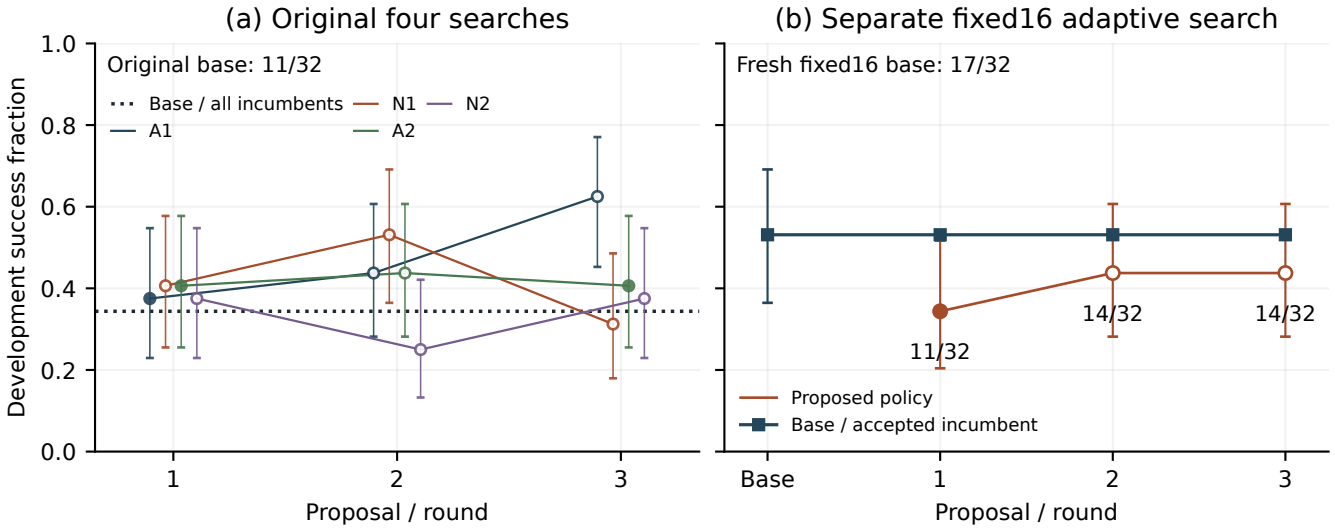


Figure 3: Development proposals (open: identity; filled: nonzero edit) and accepted incumbents, with separate baselines and nominal 95% Wilson intervals. Retained incumbents reuse their score; accepted development scores are nondecreasing by construction. Full restart panels and paired intervals accompany the blog.

bootstrap interval $[0.0625, 0.5000]$, despite zero changes: this is a cohort difference, not an edit effect. Both non-adaptive runs exhausted their activation budgets on failed captures and submitted identities. Their abstention prevents isolating adaptive reasoning under equally usable evidence. Only 24 of 368 charged logical items belonged to fully validated captures; 316 belonged to failed calls and 28 to an unchecked diagnostic. All 384 reserved candidate episodes completed.

4.3 Separate fixed16 follow-up

Offline validation passed on 31 positions from 16 legacy inputs: native repeat, identity, and packet replay were exact; instrumentation error peaked at 0.002464, with zero skips. The fresh 17/32 baseline is not an intervention estimate relative to v1.

The adaptive editor completed three evaluations in 63.43 minutes. All 80 requested activation items passed full-packet parity. It halved the projection of eight layer-17 rows onto a pregrasp success-minus-failure contrast, changing 6,225 bfloat16 values ($\|\Delta\|_2 = 0.02643$). The candidate scored 11/32 versus 17/32 base, with 5 nominal fixes and 11 breakages, and was rejected. Two identity controls each scored 14/32, with 5/8 and 4/7 fixes/breakages; they disagreed on 10/32 outcomes. The base remained incumbent. Valid capture did not yield an accepted edit or eliminate rollout variability; one run cannot establish a causal comparison with v1.

Held-out identities disagreed on 29/64 fixed16 and 37/64 original-runtime outcomes despite unchanged weights (Table 3). Differences were +0.0469 and −0.0469, with nominal intervals $[-0.1250, 0.2031]$ and $[-0.2344, 0.1406]$. Round labels share estimates; rejected edits were not substituted. The gripper control scored 36/64 versus 23/64: difference +0.2031, interval $[0.0469, 0.3594]$. Yet the extra unchanged repeat scored 35/64 versus its reused 26/64 reference: +0.1406, interval $[-0.0313, 0.3125]$. It had 21/12 nominal fixes/breakages. These correlated repeats and nominal pairing qualify the gripper contrast; its runtime adjustment is not WSP.

4.4 Repeatability and implementation limits

A post-hoc first-request audit of all 64 fixed16 identity pairs found identical joint/gripper state and seeds, but different images and actions. Median per-image mean absolute differences were 0.949/2.546 external/wrist intensity units (0–255). Of 960 planned gripper commands, 31 differed across threshold 0.5, spanning seven pairs. This audit isolates neither the source of variability nor perceptual versus motor failures.

An isolated Isaac Sim 6.0.1.0 / Isaac Lab 3.0.0b2.post1 setup used two RTX 4090s, post-bfloat16 support checks, and private RAM checkpoint staging for disk pressure. Simulator-version equivalence remains untested; reproducible setup, validation, and costs accompany the blog.

Post-freeze diagnostics completed 24/16 native forwards for v1/v2 after a separate preallocation-only correction; initial attempts failed allocation. Repeat, identity, and restoration controls were exact. All four edits changed actions on 32 first requests per runtime, each with zero binary gripper flips among 480 planned actions. V1 used new batches; v2 preserved recorded packets. These fixed-input effects do not establish closed-loop gains.

Misused condition-group names caused 62 false yellow-block wrong-object annotations across 28 original baseline episodes, including all 11 successes. Terminal success uses a separate predicate. Editors saw unvalidated logs; their influence is unknown. Both frozen runtimes retain the bug.

5 Implications and Further Discussion

Policy family and failure mechanism. We hypothesize that a localized control defect, with relevant state already represented, is easier to diagnose and patch than perceptual ambiguity or absent task competence. State-input policies can expose physical variables directly; VLAs couple visual grounding, language, and action. WAMs may expose predictive structure but also model error; these categories can overlap. Testing editability requires controlling failure type, prior training, editor access, and budget. Our single-VLA study cannot rank policy families or motor/vision repair; cross-family experiments are pending.

Why demonstrations may not transfer to robots. The original games use structured grids; the car repair uses explicit physical state. Its diffusion example recognizes generated shapes rather than improving generation [1]; the reviewed restoration loop is conjectural [22]. Thus the evidence does not establish that WSP works better for images than robotics. Mechanistically, fixed-input image evaluation can hold the stimulus constant, whereas robot actions alter observations and contacts, potentially compounding errors. Occlusion, delayed success, and reset variability complicate attribution. Games also have feedback; generators have distributed features and sampling variability. Task structure and evaluation control matter, precluding a universal modality hierarchy.

Feedback search and world models. Our loop is language-model-guided derivative-free policy search, related to RL and evolution strategies [27]; the editor remains fixed. Unlike world-model methods [28, 29], we learn no dynamics predictor. An editor's explanation is not a validated world model; learned-dynamics screening would introduce additional model error.

Limits of our evidence. Development identity repeats broke 3–10 nominal base successes against a cap of two. With only four edits, replay failures, and incorrect annotations, rejection isolates neither edit-induced harm nor a contradiction of positive studies. Physical transfer, cross-task preservation, and cost advantage remain untested.

6 Conclusion

WSP offers promising targeted repairs; current evidence does not establish routine, reliable improvement for Physical AI. Four takeaways follow:

(1) **Condition claims on the problem.** Report policy family, prior competence, failure type, and editable components; motor-versus-perception differences remain hypotheses here.

(2) **Establish measurement validity first.** Validate annotations, numerical replay, and unchanged-policy variation before attributing gains or breakages to edits.

(3) **Evaluate behavior and discovery cost.** Our loop yielded no accepted gain. Small patches establish neither cheap discovery nor preserved competence.

(4) **Advance through matched tests.** Reproduce positive cases, calibrate gates prospectively, and evaluate independently. Retain failures and implementation costs.

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